

AI LinkedIn Content Automation

An n8n pipeline that researches a topic, drafts two LinkedIn/X angles with AI, sends both for Telegram approval, and publishes the chosen post to LinkedIn automatically.



1 Executive Summary

Consistent LinkedIn posting takes real research time, and a purely automated poster risks publishing something off-brand. This workflow closes that gap: it pulls one queued topic at a time, has Firecrawl find real sources, and has Groq draft two distinct LinkedIn/X angles with matching AI images — then sends both drafts to Telegram for a one-tap decision. Nothing reaches LinkedIn until a person picks A or B; Regenerate and Skip are always available instead.

2 Key Metrics

Trigger	Daily 9AM schedule, or a manual start, reading one queued topic at a time
Research Engine	Firecrawl finds sources; Groq (gpt-oss-120b) drafts two distinct angles from them
Image Generation	Cloudflare Workers AI renders a matching preview image for each angle
Draft Validation	Both angles must include a real source URL and stay within a 280-character X limit
Approval Channel	Telegram sends both drafts with images; Publish A/B, Regenerate or Skip are the only actions
Publishing	The chosen post and image are published to LinkedIn only after a tap on A or B
Callback Security	Every button tap is matched against the chat ID, topic ID and a numeric review ID

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DRAFT ANGLES PER TOPIC

280

X CHARACTER LIMIT

0

AUTO-PUBLISHED POSTS

3 Business Value

- ✓ **One topic queue, zero blank-page problem** — add a topic_id and topic to a sheet; the workflow researches and drafts it.
- ✓ **Every draft is source-backed** — Groq must cite a real Firecrawl URL, or the draft is rejected before reaching Telegram.
- ✓ **Full creative control from a phone** — two distinct angles and images arrive in Telegram; approve, regenerate, or skip.
- ✓ **Nothing publishes without a tap** — LinkedIn is only touched after Publish A or Publish B is explicitly chosen.

4 Architecture Overview

Three connected flows, all reading and writing the same Topics tracking sheet.

FLOW	TRIGGER	PURPOSE
Research & Draft	Daily 9AM schedule, or manual start	Picks one queued topic, researches it, and drafts two angles with images for review
Telegram Review	Telegram trigger, on button tap	Validates the tap and routes it to Publish, Regenerate, or Skip
Publish & Record	Continuation of an approved tap	Downloads the chosen image, publishes to LinkedIn, and logs the result

5 How It Works

Research & Draft	Telegram Review	Publish & Record
1 One queued topic is picked from the sheet and marked researching so it can't be picked twice. 2 Firecrawl finds real sources on the topic; the results are cleaned into a source list. 3 Groq drafts exactly two angles — LinkedIn copy, X copy, an image prompt, and cited sources for each. 4 Each draft is validated (real source URL, X under 280 chars) before Cloudflare renders preview images.	1 The bot sends both angles with their preview images and Publish A / B / Regenerate / Skip buttons. 2 A button tap is matched against the chat ID, topic ID and a numeric review ID before it's trusted. 3 The matching sheet row must still be awaiting_approval, or the tap is silently ignored. 4 The validated action is routed: A/B to publishing, R to regenerate, S to skip.	1 The chosen row is reserved and the selected post's text and image prompt are prepared. 2 The approved preview image is downloaded from Telegram as a file. 3 The post and image are published to LinkedIn under the configured profile. 4 The LinkedIn URL is logged; the X copy is sent to Telegram for manual posting.

6 Reliability & Security Design

- ◆ **Draft validation before Telegram** — Groq's JSON must contain exactly two options, each with a real cited source URL.
- ◆ **X length is enforced in code** — any draft over 280 characters is rejected, never silently truncated.
- ◆ **Callback tap is re-verified** — chat ID, topic ID and a numeric review ID must all match before an action is trusted.
- ◆ **Stale taps are ignored** — if the sheet row has moved on from awaiting_approval, a late button press does nothing.
- ◆ **LinkedIn is never touched by Regenerate or Skip** — only Publish A or Publish B reaches the publishing branch.
- ◆ **One topic in flight at a time** — the workflow won't start a new topic while one is researching, publishing, or awaiting approval.

7 Setup Checklist

- ☒ Set the Google Sheet ID, Telegram chat ID and Cloudflare account ID in both Config nodes.
- ☒ Connect separate API credentials for Firecrawl, Groq, Cloudflare and Telegram.
- ☒ Select your personal profile in the Publish LinkedIn node.
- ☒ Add a topic to the Topics sheet with a unique topic_id and status=queued to start a run.
- ☒ Test one full cycle — research, Telegram review, and a Publish A/B tap — before scheduling it daily.

8 Node-by-Node Map

Research & Draft	Read topic queue → Pick one topic → Mark researching → Firecrawl search → Clean research → Groq two angles → Validate two drafts
Image & Preview	Cloudflare Image A/B → Binary Image A/B → Preview Image A/B → Preview Text A/B → Save drafts → Choose A or B
Telegram Review	Telegram choices → Validate callback → Find review row → Check pending choice → Route decision
Publish & Record	Reserve chosen row → Prepare selected post → Get a file → Publish LinkedIn → Build publishing record → Save LinkedIn success → Send X copy and LinkedIn link

SUMMARY

Designed and built end-to-end in n8n, demonstrating source-grounded AI drafting, dual-angle content generation with matching images, and a Telegram-approved LinkedIn publishing loop.